

How a Memory Forms

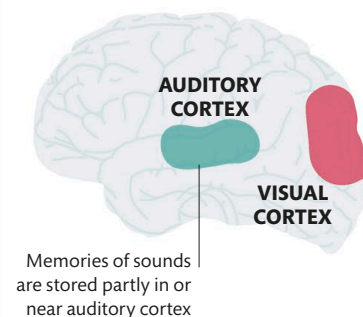
When networks of neurons in the brain are repeatedly activated, changes in the cells strengthen their connections, making it easier for each to activate the next (see pp.26–27). This process is known as long-term potentiation.

Strengthening connections

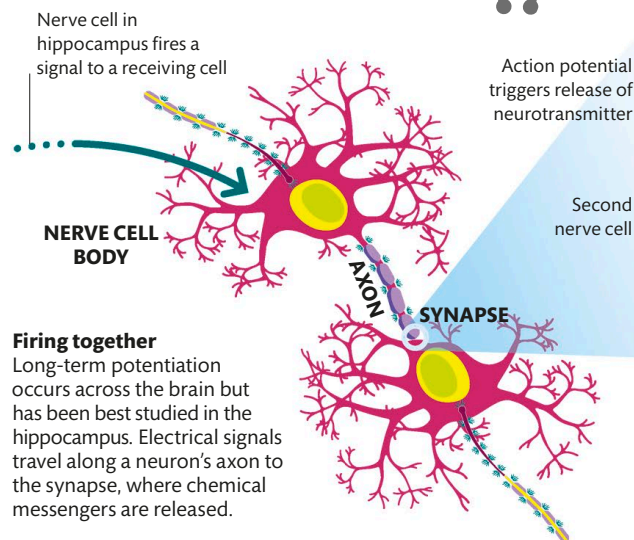
When you repeatedly activate a group of neurons—by practicing a skill or revising facts, for example—they begin to change. This is how we form long-term memories (see p.135) in a process called long-term potentiation, which depends on various mechanisms taking place in brain cells. The first (presynaptic) neuron makes more neurotransmitters release when the signal reaches it, and the second inserts more receptors into its membrane. This speeds up transmission at the synapse. Something like driving a car, which seems complex when you start, can become effortless as the neural pathways involved become more efficient. If this paired activation is repeated enough, new dendrites can grow, linking the two neurons via new synapses, giving the message alternative pathways and helping it travel even faster.

MEMORY TRACES

Scientists have recently been able to pinpoint a precise memory trace in someone's brain. In general, memories tend to be stored near the area of the brain that relates to how they were formed. For example, memories for voices would be near the language centers, and things that you have seen are stored, at least partly, near the visual cortex.

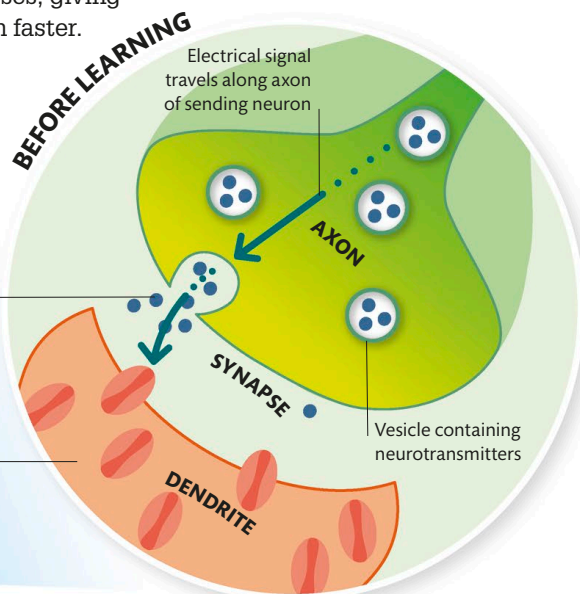


MORE THAN 100 DIFFERENT NEUROTRANSMITTERS HAVE BEEN IDENTIFIED



Firing together

Long-term potentiation occurs across the brain but has been best studied in the hippocampus. Electrical signals travel along a neuron's axon to the synapse, where chemical messengers are released.



- 1 Before learning, only a weak connection exists between neurons. One action potential (pulse of electrical current) from the first cell releases only a small amount of neurotransmitters, and this may or may not be enough to activate the next neuron, which has just a few receptors.